

Adverse Event Safety Signal Monitoring: Project Case Study

Executive Summary

Continuous adverse event (AE) monitoring is essential during clinical trials to ensure patient safety and detect potential drug induced harm early. This project provides a statistical safety analysis of the publicly available CDISC SDTM Pilot Project dataset (dm.xpt with 306 subjects and ae.xpt with 1,191 AE records).

The study evaluates safety outcomes across three active trial arms Placebo (n=86), Xanomeline Low Dose (n=84), and Xanomeline High Dose (n=84), explicitly excluding 52 Screen Failures from all calculations. The end-to-end pipeline uses SQL for relational aggregations, R for statistical significance and disproportionality signal detection, and Power BI to deliver an interactive two-page safety dashboard.

Tools and Technologies

Tool	Role
SQL (SQLite)	Aggregating raw AE and demographic records into incidence, severity, and system organ class summaries by treatment arm
R	Statistical significance testing (chi-square) and disproportionality signal detection (Proportional Reporting Ratio)
Power BI	Two page interactive safety dashboard translating the statistical findings into a stakeholder facing view

Key Insights & Business Impact

- **Dose-Dependent Incidence:** AE incidence scaled directly with dose 94.0% of High Dose subjects experienced at least one AE, compared to 91.7% in Low Dose and 80.2% in Placebo.
- **Statistical Significance:** A chi-square test confirmed that the High Dose vs. Placebo incidence difference was statistically significant ($p = 0.014$), whereas the Low Dose vs. Placebo difference was borderline ($p = 0.055$).
- **Severity & Serious Adverse Events (SAEs):** Severe AEs were reported in 10 High Dose events, 25 Low Dose events, and 8 Placebo events. Additionally, 3 total SAEs were identified (2 in High Dose, 1 in Low Dose, 0 in Placebo).
- **Pharmacovigilance Signal Detection (PRR):** A Proportional Reporting Ratio (PRR) screening (threshold: $PRR > 2.0$, $n \geq 3$) flagged 6 specific symptoms disproportionately reported in the High Dose arm:

AETERM	n_test_arr	n_placebo	PRR
APPLICATION SITE ERYTHEMA	23	3	5.07
DIZZINESS	18	3	3.97
SINUS BRADYCARDIA	12	2	3.97
NAUSEA	13	3	2.87
APPLICATION SITE PRURITUS	35	10	2.32
PRURITUS	38	11	2.29

Approach

Step 1: Relational Data Aggregation (SQL)

- Imported native SAS Transport (.xpt) files via Pandas and R.
- Filtered out Screen Failure subjects to avoid diluting safety denominators.
- Aggregated AE incidence rates, severity distributions (Mild, Moderate, Severe), top System Organ Classes (SOC), and SAE counts by treatment arm.

Step 2: Hypothesis Testing & Pharmacovigilance Signals (R)

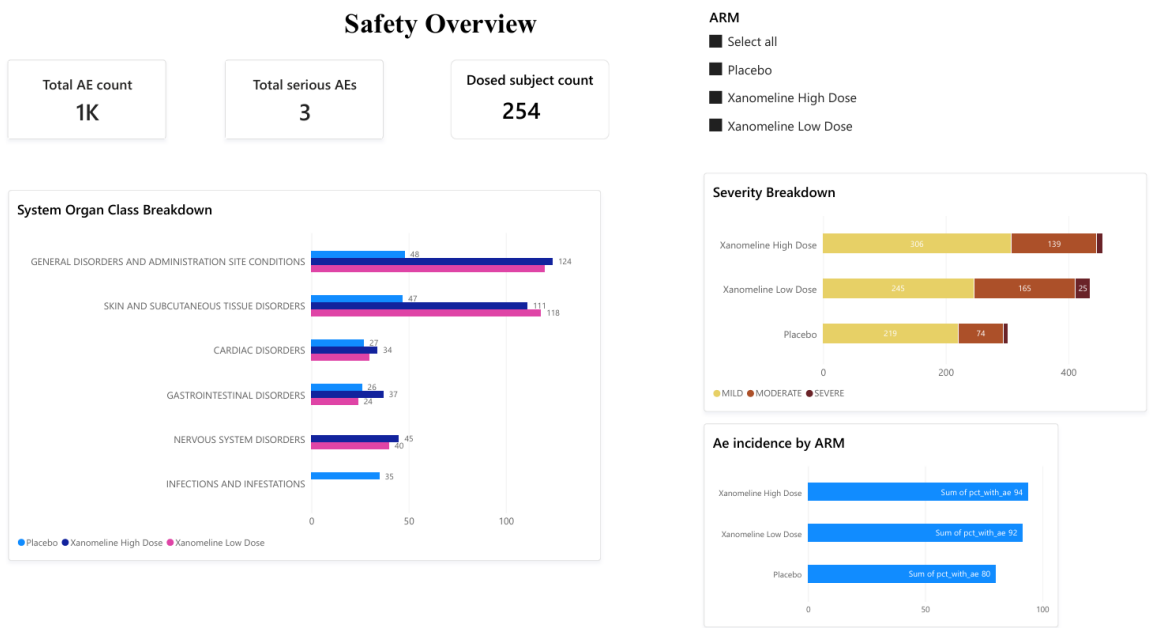
- Executed Chi-Square tests of independence comparing AE incidence rates between treatment arms.

- Calculated Proportional Reporting Ratios (PRR) for the top 15 reported terms using industry-standard FDA FAERS screening rules (PRR > 2.0, n ≥ 3 cases).

Step 3: Interactive Dashboard (Power BI)

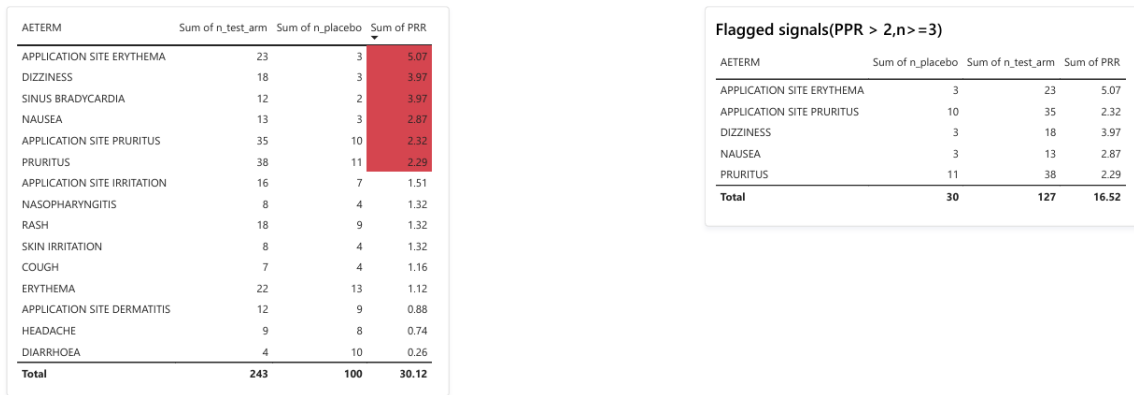
Constructed a two-page safety dashboard:

- Safety Overview Page:** High-level KPI cards, incidence rates, and severity distribution charts.



- Signal Detection Page:** Conditional formatted PRR grid highlighting safety signals above the screening threshold.

Signal Detection



Impact / Results

This pipeline produces two concrete, reusable deliverables:

- **A prioritized, quantified list of specific symptoms** worth flagging to the clinical team for further review rather than 1,191 individual AE records that no one has time to read one by one.
- **A dashboard any non-technical stakeholder can open** and immediately see incidence, severity, and the flagged signals without needing to run a single query themselves.